Francesco Innocenti

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My research focuses on understanding the training dynamics and scaling behaviour of artificial and biological neural networks. I like to understand things through the lens of optimisation theory, and am broadly interested in energy-based models, time-series forecasting, and all things AI-related.

Experience

Nov 2025 - present **Postdoctoral Researcher**, *University of Oxford*, UK.
- Researching the scaling and optimisation dynamics of brain-inspired learning algorithms.

Oct 2023 - Apr 2024 **Applied Scientist Intern**, *Amazon*, Barcelona.
- Helped improve and evaluate a short-term forecast of Amazon packages delivered throughout Europe, contributing to an internal conference paper and $MM savings in operational costs.

Education

Sept 2021 - 2025 **PhD, Machine Learning & Theoretical Neuroscience**, *University of Sussex*, UK.
- Thesis: "Towards Scaling Deep Neural Networks with Predictive Coding: Theory and Practice".
- Teaching Assistant on "Fundamentals of ML" MSc course.

Sept 2018 - Jun 2021 **BSc, Psychology with Cognitive Neuroscience**, *Goldsmiths, University of London*.
- 1st Class Honours.†
- Thesis: "Modelling the Evolution of Visual Perception with Evolutionary Algorithms".

Open-source contributions

- Developed JPC (★ 81), a JAX library for training neural networks with predictive coding.
- Maintaining a repository of papers on the Maximal Update Parameterisation (μP).
- Curated repositories of papers on Neuro-AI (★ 52) and the Hessian of neural networks (★ 2).

Skills

Coding	Python (highly experienced), git (highly experienced), AWS (experienced).
Autodiff.	PyTorch and JAX (both highly experienced).
Languages	English (proficient), Italian (native), Spanish (fluent).

Selected papers

[1] **Innocenti, F.**, Achour, E. M., and Rafal Bogacz (2026). On the Infinite Width and Depth Limits of Predictive Coding Networks. *(Under review.)*

[2] **Innocenti, F.**, and Achour, E. M. (2025). A Simple Generalisation of the Implicit Dynamics of In-Context Learning. *NeurIPS Workshop on What Can('t) Transformers Do?.*

[3] **Innocenti, F.**, Achour, E. M., and Buckley, C. L. (2025). μPC: Scaling Predictive Coding to 100+ Layer Networks. *Advances in Neural Information Processing Systems 38.*

[4] **Innocenti, F.** et al. (2024). Only Strict Saddles in the Energy Landscape of Predictive Coding Networks? *Advances in Neural Information Processing Systems 37.*

Selected awards

- $25,000 in Google Cloud credits awarded by the Google TPU Research Cloud programme.
- Best Paper Award at the ICML 2023 Workshop on Localized Learning.
- †British Psychological Society (BPS) Award for highest performance in undergraduate degree.